

CS & IT ENGINEERING



Operating System

Process Synchronization

Lecture - 2

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Recap of Previous Lecture

doubt?

Topic

Synchronization

Topic

Race Condition



Topics to be Covered



Topic

Critical Section

Topic

Solution of Critical Section Problem

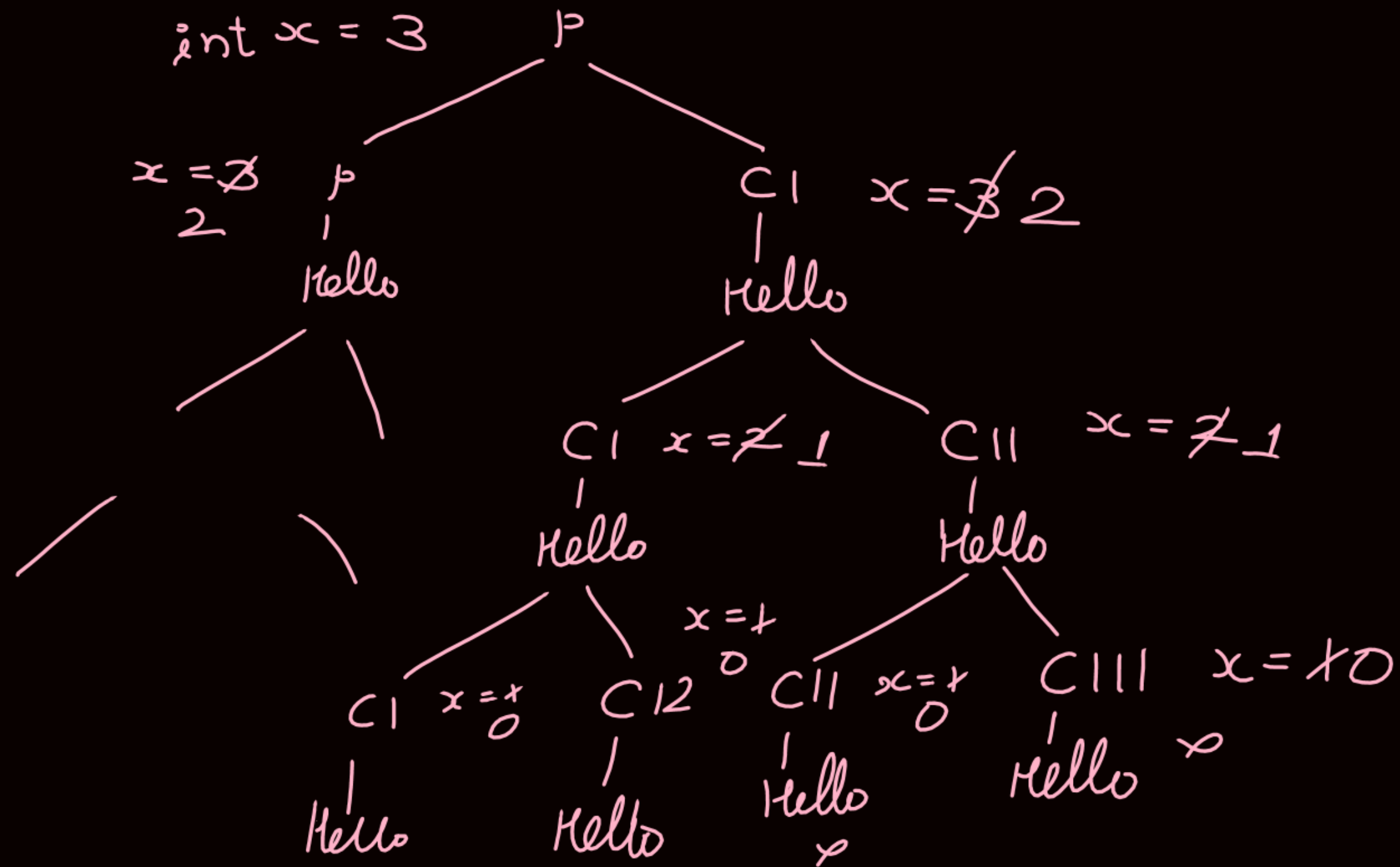
#Q. Consider the following code snippet using the `fork()` and `wait()` system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;  
while(x > 0) {  
    fork();  
    printf("hello");  
    wait(NULL);  
    x--;  
}
```

Ans = 14

The total number of times the `printf` statement is executed is 14?

GATE-2024



#Q. The following pair of processes share a common variable X.

Process A	Process B
int Y;	int Z;
$Y = X * 2;$	$Z = X + 1;$
$X = Y;$	$X = Z;$

$$X = \cancel{4} 9$$

9, 10, 8, 5

X is set to 4 before either process begins execution. As usual, statements within a process are executed sequentially, but statements in process A may execute in any order with respect to statements in process B. How many different values of X are possible after both processes finish executing?

[NAT]

The following two functions $P1$ and $P2$ that share a variable B with an initial value of 2 execute concurrently.

P1()	P2()
$\{$ $C = B - 1;$ $B = 2 * C;$ $\}$	$\{$ $D = 2 * B;$ $B = D - 1;$ $\}$

3, 4, 2

The number of distinct values that B can possibly take after the execution is 3.

[NAT]

Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100.

$$① = 100 - 50 + 20 = 70$$

$$① = 100 - 20 + 10 = 90$$

✓ P1	✓ P2	P3
⋮	⋮	⋮
$D = \overset{100}{D} + 20$	$D = \overset{100}{D} - 50$	$D = \overset{120}{D} + 10$
⋮	⋮	⋮
⋮	⋮	⋮

The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of $Y - X$ is 80.

$$X = 50$$

$$Y = 130$$

$$Y - X = 80$$

Ques all possible distinct values of D in last question?

Ans

$$100 + 20 - 50 + 10$$

120

50

110

70

130

60

80

Ques) 4 processes

$$x = 1$$

P1	P2	P3	P4
$x = x + 1$	$x = x + 2$	$x = x - 4$	$x = x - 8$

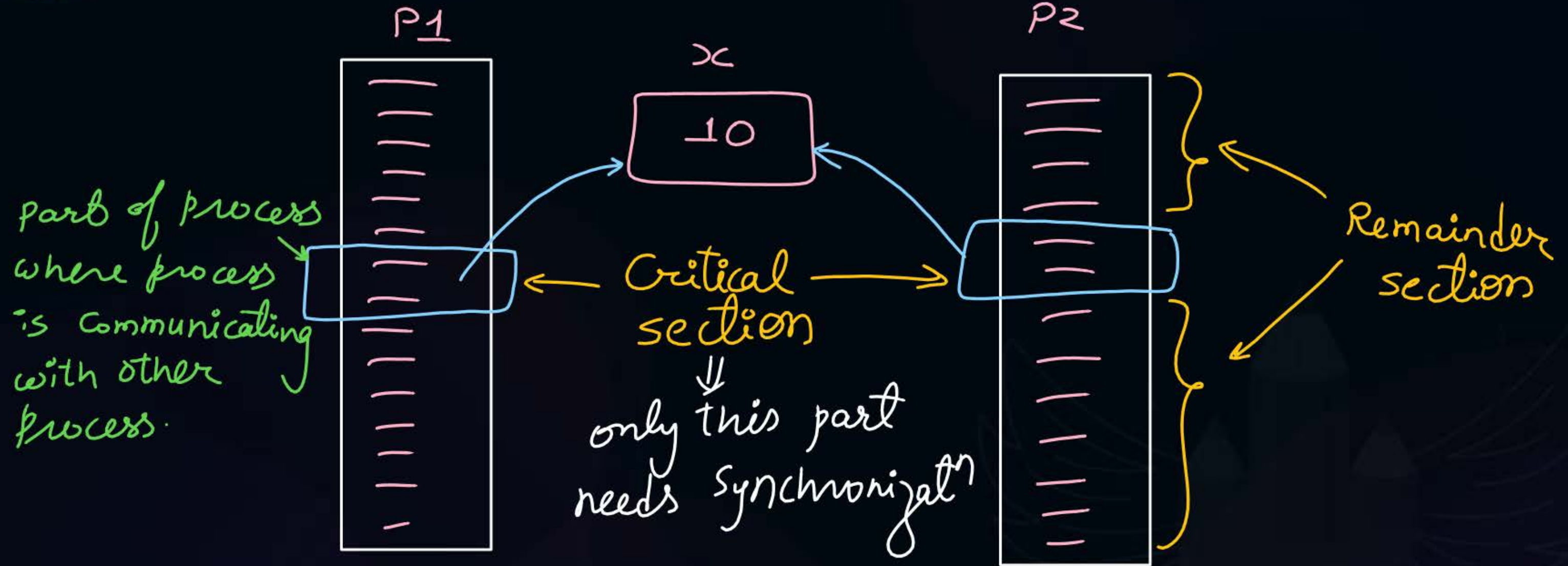
$$1 + 1 + 2 - 4 - 8$$

15 distinct
values

2	4	0	-8
3	-2	-4	
-3	-6	-10	
-7	-1	-9	
	-5		
	-11		



Topic : Entire Process Requires Synchronization?





Topic : Critical Section



The critical section is a code segment where the shared variables can be accessed.

↳ Communication

Critical section problem:- providing synch. for critical sections



Topic : Solution of Critical Section Problem

Requirements of Critical Section problem solution:

1. Mutual Exclusion
 2. Progress
 3. Bounded Waiting
- must*
- Perfect*



Topic : Solution of Critical Section Problem

Mutual Exclusion:

If one process is executing the critical section, then other process is not allowed to enter into critical section.

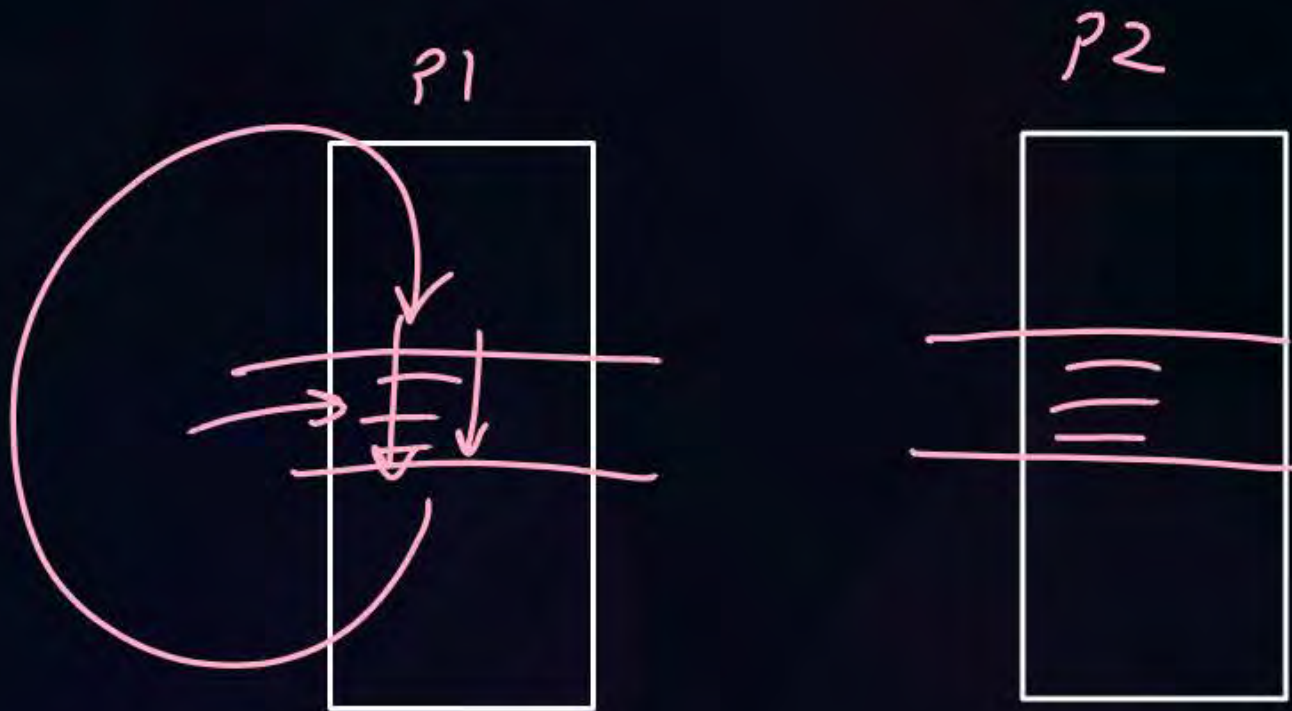




Topic : Solution of Critical Section Problem

Progress:

If no any process is in critical section and any process wants to enter into critical section, then the process must be allowed.





Topic : Solution of Critical Section Problem

Bounded Waiting:

If a process p_1 is executing in critical section and other process p_2 is waiting for critical section, then the waiting time of p_2 must be bounded. Which means p_1 must not enter in to critical section again and again by keeping p_2 in waiting for long.



Topic : 2-Process Solution

Solution

Entry section

C.S.

← responsible for
implementing 3 requirements

Exit section

← announcement
of a process that
it is leaving C.S.



2 mins Summary

Topic

Critical Section

Topic

Solution of Critical Section Problem





Happy Learning

THANK - YOU

